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Comparison of Skyfield SGP4 and Basilisk Numerical Orbit Propagation for GNSS-R Formation-Flying Applications

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ABSTRACT

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CHAPTER

ONE

NOTES

Strongest narative for my thesis:

1. My specialization project showed that single-TLE SGP4 is not sufficient as ground truth, and that simplified drag limited the conclusions.
2. The Master's thesis improves upon the validation reference by using regularly update TLEs, making Skyfield-SGP4 a short-horizon, observation-updated reference
3. Basilisk is then validated as sufficiently accurate for long time-horizon mission simulation
4. The validated Basilisk propagator is used inside a realistic two-satellite GNSS-R formation simulator
5. Mission lifetime is estimated from fuel depletion + formation loss, with battery/power constraints shaping operations.

Scope: Focus on:

- Basilisk propagation validation
- Two-satellite deployment and formation acquisition
- Autonomous mode switching
- battery and fuel lifetime curves
- formation evolution
- mission lifetime estimate

Include, but keep on a high level:

- GNSS-R signal physics (just what is required to enable observation)
- Advanced controller derivations (both attitude and formation)

CHAPTER

TWO

INTRODUCTION

https://www.itk.ntnu.no/emner/masteroppgave

This has to be included in the report to make it clear to the sensors what has been done:

- Information, software and equipment available/used
- What help has been received during the work
- The use of AI. The AI-declaration schema MUST be inserted on the first page after the title page

2.0.1 FNs sustainability goals

2.0.2 Background and motivation

2.1 Take-aways from the specialization project

2.1.1 Research problem

2.1.2 Research objectives

2.1.3 Research questions

CHAPTER

THREE

LITERATURE REVIEW

The literature review conducted for this paper can be divided into three themes:

- GNSS-R Small-Satellite Missions
- Formation Flying and Formation Maintenance for Small Satellites
- Integrated Satellite Mission Simulation and Lifetime Analysis

3.1 GNSS-R Small-Satellite Missions

- Scientific evolution:
  1. GNSS as a signal of opportunity
  2. GNSS-R feasibility for different remote sensing applications. Bistatic radar principle
  3. Early demonstrations of GNSS-R remote sensing on aircraft mission
  4. Spacecraft demonstrations
  5. Validation of technology for different applications
  6. Expansion of GNSS-R Mission applications (include jamming/spoofing detection)
- Mapping useful GNSS-R missions after validation of technology at spacecraft altitudes
- Feasibility of cooperative swarms and their respective formation types w. benefits

Reflected Global Navigation Satellite Systems (GNSS) signals were first discussed as a

3.2 Formation Flying and Formation Maintenance for Small Satellites

3.3 Integrated Satellite Mission Simulation and Lifetime Analysis

3.4 Research gaps and thesis contributions

CHAPTER

FOUR

BACKGROUND THEORY

4.1 Reference Frames

A reference frame $\mathcal{F}^{rf} : \left\{\mathcal{O}_{rf};\ \hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\right\}$ is uniquely defined by three dextral orthogonal unit vectors $\left\{\hat{\mathbf{x}}_{rf}, \hat{\mathbf{y}}_{rf}, \hat{\mathbf{z}}_{rf}\right\}$ and the origin $\mathcal{O}_{rf}$ [1]. This section will present rigid mathematical definitions for the most important reference frames used in the paper. The individual reference frame indexes are assigned to be consistent with the conventional Basilisk notation.

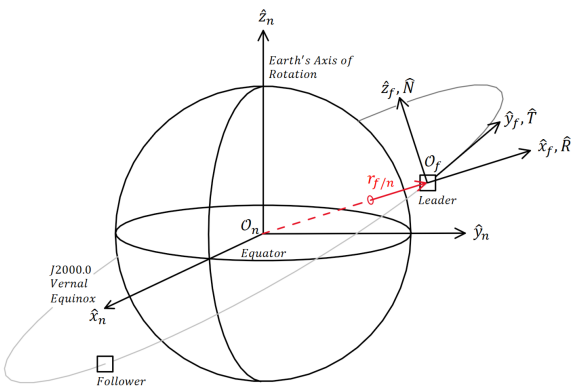

Figure 4.1.1: Illustration of the ECI ($\mathcal{F}^n$)and RTN ($\mathcal{F}^f$) reference frame definitions. The basis unit vectors are displayed with different lengths for illustrative purposes. Adopted from [2, 3] and modified from [4]

.

4.1.1 J2000 Earth-Centered Inertial (ECI) Frame

The Earth-Centered Inertial (ECI) reference frame, $\mathcal{F}^{rf} : \{\mathcal{O}_n;\ \hat{\mathbf{x}}_n, \hat{\mathbf{y}}_n, \hat{\mathbf{z}}_n\}$, is fixed relative to the distant stars, and is therefore commonly approximated as an inertial frame in which Newtonian mechanics are valid [5]. The frame is defined with its origin in the Earth's Center of Mass (CM), with $\hat{\mathbf{z}}_n$ pointing along its rotational axis towards the geographical North Pole, $\hat{\mathbf{x}}_n$ pointing towards the vernal equinox through the equator, and $\hat{\mathbf{y}}_n$ defined according to the right-hand rule. Multiple variants of the ECI definition exists, primarily varying in the epoch used to define the Earth's equatorial plane and equinox direction. In this thesis, the J2000 ECI frame is adopted. It is defined using the J2000.0 reference epoch, equivalent to 1st January 2000, at 12:00 Terrestrial Time (TT) [6]. The frame definition is illustrated in Figure 4.1.1.

4.1.2 Earth-Centered Earth-Fixed (ECEF) Frame

$\mathcal{F}^e : \{\mathcal{O}_e;\ \hat{\mathbf{x}}_e, \hat{\mathbf{y}}_e, \hat{\mathbf{z}}_e\}$

4.1.3 Radial-Transverse-Normal (RTN) Frame

The Radial-Transverse-Normal (RTN) reference frame, $\mathcal{F}^f : \left\{\mathcal{O}_f;\ \hat{\mathbf{x}}_f, \hat{\mathbf{y}}_f, \hat{\mathbf{z}}_f\right\}$, is a local orbital frame that is useful for expressing relative motion between spaceborne objects. Vallado presents the frame as the *Satellite coordinate system, RSW* in [7], but the *RTN* name is chosen for this thesis to better convey each component's physical significance. For use in formation flight, the coordinate frame origin is assigned to the leader satellite, l's CM. The dextral orthogonal unit vectors are defined with the $\hat{\mathbf{x}}_f$ pointing in the radial direction along the satellite's Earth-relative position vector, the $\hat{\mathbf{z}}_f$ axis points normal to the orbital plane spanned by the position and velocity vector, and $\hat{\mathbf{y}}_f$ fulfills the right-hand rule. Note that the latter will always point in direction of positive velocity, but will only be perpendicular to the velocity for circular orbits [7, 8]. Using this definition, the frame can be expressed mathematically as:

$$\hat{\mathbf{R}}^n = \hat{\mathbf{x}}_f^n = \frac{\mathbf{r}_{l/n}^n}{||\mathbf{r}_{l/n}^n||}, \qquad \hat{\mathbf{N}}^n = \hat{\mathbf{z}}_f^n = \frac{\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n}{||\mathbf{r}_{l/n}^n \times \mathbf{v}_{l/n}^n||}, \qquad \hat{\mathbf{T}}^n = \hat{\mathbf{y}}_f^n = \hat{\mathbf{z}}_f^n \times \hat{\mathbf{x}}_f^n \tag{4.1}$$

Here, $r_{l/n}^n$ and $v_{l/n}^n$ are used to denote the Earth-relative leader position and velocity expressed in ECI, while $\hat{\mathbf{y}}_f$ and $\hat{\mathbf{z}}_f$ correspond to the along- and cross-track directions, respectively. An illustration of the leader satellite's position relative to $\mathcal{O}_n$ and its affixed RTN frame is shown in Figure 4.1.1.

4.1.4 Body frame

$\mathcal{F}^b : \{\mathcal{O}_b;\ \hat{\mathbf{x}}_b, \hat{\mathbf{y}}_b, \hat{\mathbf{z}}_b\}$

4.1.5 Attitude Reference Frame

$\mathcal{F}^r : \{\mathcal{O}_f;\ \hat{\mathbf{x}}_r, \hat{\mathbf{y}}_r, \hat{\mathbf{z}}_r\}$

4.2 Modified Rodrigues parameters

4.2.1 Classic Orbital elements

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4.7 GNSS-R operations

4.7.1 Pointing requirements

4.7.2 Formation constraints

4.8 Simplified General Perturbations 4 (SGP4)

CHAPTER

FIVE

METHODOLOGY

- Should include a mass budget table similar to 'Table 4-4: SSTL-Cube Mass Budget Breakdown per Sub-System' in the HYDROSWARM paper

CHAPTER

SIX

RESULTS

Plotting ideas for part 2:

- The big important plot: Fuel level over time
- Each subsystem should be validated in addition to showing the full fuel consumption over simulation duration:
  - RW speed/torque step response
  - Thruster force output as a function of fuel level (/tank pressure)
  - MRP feedback controller tracking error and/or step response (would also be interesting to see a step response for different fuel levels)
  - Decomposed net power and battery level (should also show the illumination factor in the same/adjacent plot and the FSW state)

CHAPTER

SEVEN

DISCUSSION

IMPORTANT!

I have done some initial testing using the Basilisk Station Keeping example, and it seems like a strict CPO or Concentric formation is unachievable for the spacecraft reconfig module. If I have time, I should absolutely implement a HCW/ROE based controller for formation control. If I don't have time, I should just argue that this should be expanded in the future, and make a solid discussion about how my fuel consumption estimate is in comparison to the more realistic HCW case.

CHAPTER

EIGHT

CONCLUSIONS

8.1 Conclusion

REFERENCES

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[3] Sascha Metz. "Implementation and Comparison of Data-Based Methods for Collision Avoidance in Satellite Operations". Master's Thesis. Darmstadt, Germany: Technische Universität Darmstadt, 2020. URL: https://www.researchgate.net/publication/344268107_Master_Thesis_Implementation_and_comparison_of_data-based_methods_for_collision_avoidance_in_satellite_operations.

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[5] Olav Egeland and Jan Tommy Gravdahl. *Modeling and Simulation for Automatic Control*. Trondheim, Norway: Marine Cybernetics, 2002. ISBN: 82-92356-01-0.

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[7] David A. Vallado. *Fundamentals of Astrodynamics and Applications*. 2nd ed. Vol. 12. Space Technology Library. El Segundo, CA: Microcosm Press and Kluwer Academic Publishers, 2001. ISBN: 9781881883128.

[8] Sean R. Stratton. "Benefits of RTN Reference Frame as Standard for Optical Observation Data". In: *Advanced Maui Optical and Space Surveillance Technologies Conference (AMOS)*. Approved for Public Release, Distribution Unlimited. SciTec, Inc. 2025. URL: https://amostech.com/TechnicalPapers/2025/Astrodynamics/Stratton.pdf.

APPENDICES

A - GITHUB REPOSITORY

All simulator code and LaTeX-files used in this document are included in the GitHub repositories linked below.

Specialization project report LaTeX repository link

- https://github.com/chystad/Specialization-Project-Report-LaTeX-

Orbit Simulator repository link

- https://github.com/chystad/Orbit_Simulator

B - NUMERICAL VALUES

Numerical Values for Physical Constants

$Constant$	$Value$	$Unit$
J_0	-1	-
J_1	0	-
J_2	1.08263×10^{-3}	-
J_3	-2.53266×10^{-6}	-
J_4	-1.61962×10^{-6}	-
P_{Sun}	$4.578299965 \times 10^{-6}$	Ws/m^3
$1AU$	149597871	km
R	6378136.6	m
H_0	7200	m
ρ_0	1.225	kg/m^3
$-$	$-$	$-$
$-$	$-$	$-$
$-$	$-$	$-$

Table .0.1: Physical constants and associated numerical value. J-terms courtesy of